

Technical Report: Prescriptive Analytics & Optimization

Implementation of Linear Programming for Logistics Efficiency

1. Executive Summary

This report outlines the development of a prescriptive analytics tool designed to optimize delivery routes using Python. By leveraging Linear Programming (LP), the model provides a mathematically guaranteed "best" course of action for moving goods from three depots to three retail stores. This project transitions traditional manual logistics planning into a modern, automated workflow.

2. Professional Context & Strategic Advancement

"Prescriptive analytics represents the final stage of business analytics... moving beyond telling us what will happen to telling us what we should do to make it happen." — ACCA Technical Article

Leading professional bodies such as **ACCA** and **CIMA** emphasize that the modern accountant and business leader must move beyond descriptive data. Automated business decision-making is no longer a luxury but a strategic necessity.

The **World Economic Forum (WEF)** identifies "Analytical Thinking and Innovation" as a top skill for 2025. Automated decision systems, like the one implemented here, reduce human error and allow for immediate data availability in complex workflows, ensuring that logistics networks can adapt to supply chain shocks in real-time.

3. Technical Implementation

Libraries Utilized

- **SciPy (scipy.optimize.linprog)**: The mathematical core used to solve the minimization problem via the HiGHS solver.
- **NumPy**: Essential for high-performance matrix operations and data flattening.
- **Pandas**: Used for structuring results into human-readable business tables.

- **Streamlit:** The UI framework that turns the script into a web-accessible dashboard.

Core Optimization Logic (Snippet)

```
# Optimization via SciPy HiGHS Solver
res = linprog(
    c=cost_vector,
    A_ub=capacity_matrix,
    b_ub=store_capacities,
    A_eq=supply_matrix,
    b_eq=depot_inventory,
    method="highs"
)

# Results displayed as integers for business reporting
optimal_schedule = res.x.reshape(3, 3).astype(int)
```

4. The Role of GenAI & Human Synergy

This project was developed with **GenAI assistance**, illustrating the emerging "Co-Pilot" workflow. The human provides the strategic domain knowledge (understanding the logistics problem and constraints), while the AI assists in the rapid prototyping of boilerplate code, UI styling, and documentation. This synergy accelerates the development lifecycle, allowing business analysts to build production-ready tools in hours rather than weeks.

5. Advantages of the Python Ecosystem

Python offers **immediate data for workflows**. Unlike legacy spreadsheets, Python-based solutions can be integrated into live APIs, connected to real-time warehouse databases, and scaled to handle thousands of depots. The ability to deploy via **Streamlit** democratizes data science, putting complex prescriptive models directly into the hands of operational managers.